

Symmetry

A symmetric distribution has approximately matching shapes on both sides of its center.

What would symmetry look like in a histogram?

Unimodal and bimodal distributions

Unimodal distributions have one mode or peak; bimodal distributions have two.

Can a distribution be symmetric and bimodal?

Right skew

Synthetic data. A right-skewed distribution has a longer tail toward larger values.

Which observations form the tail in 1, 2, 2, 3, 3, 4, 12?

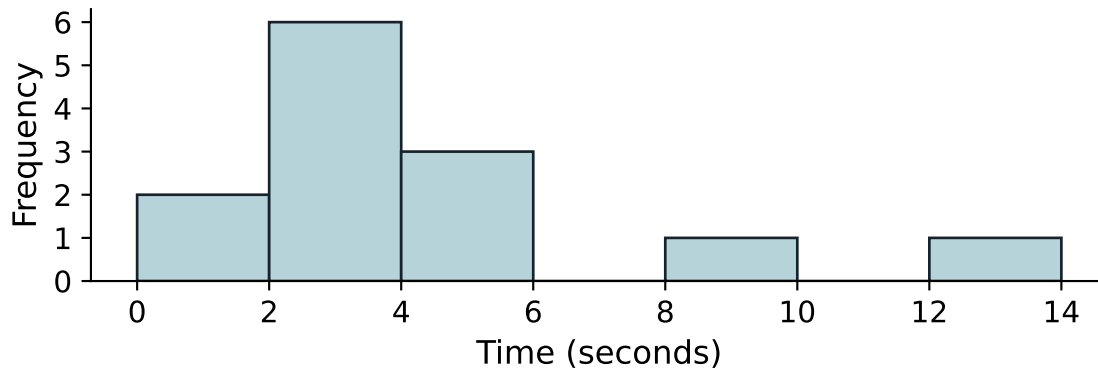
Left skew

Synthetic data. A left-skewed distribution has a longer tail toward smaller values.

Which observations form the tail in 1, 8, 9, 9, 10, 10, 11?

Read a right tail

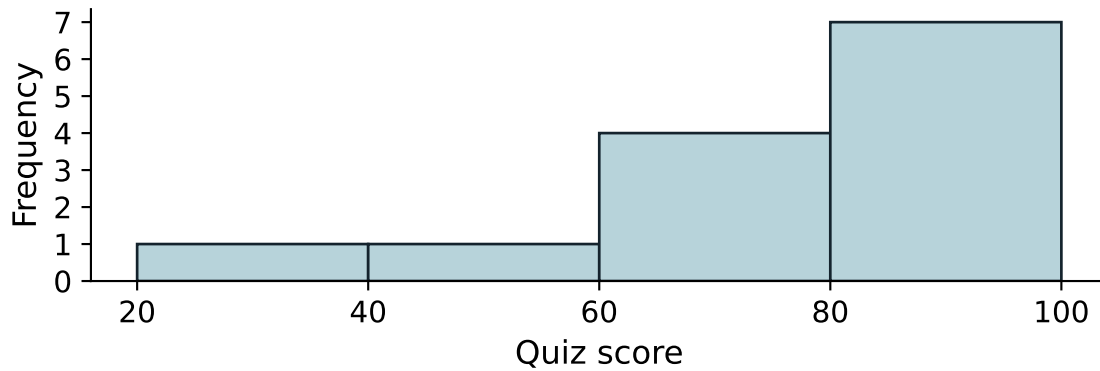
Synthetic times; bins are left-inclusive.



In which direction is the distribution skewed?

Read a left tail

Synthetic quiz scores.



In which direction is the distribution skewed?

Mean and a right tail

Synthetic values: 1, 2, 2, 3, 3, 4, 12.

Is the mean above or below the median?

Mean and a left tail

Synthetic values: 1, 8, 9, 9, 10, 10, 11.

Is the mean above or below the median?

A symmetric example

Synthetic values: 2, 3, 3, 4, 5, 5, 6.

How do the mean and median compare?

A larger extreme

Synthetic values: 1, 2, 2, 3, 3, 4, 12. Replace 12 with 40.

How does the mean move relative to the median?

Skew names the tail

A histogram has most observations on the left and a few far to the right.

Is it left-skewed or right-skewed?

The mode and shape

Synthetic values: 1, 2, 2, 2, 3, 4, 12.

Where are the mode, median, and mean relative to one another?

Comparing dot plots and centers

Synthetic groups: A: 1, 2, 2, 3, 3, 4; B: 1, 2, 2, 3, 3, 12.

How do dot plots help explain the difference in their means and medians?

A common pattern

For many unimodal right-skewed distributions, the mean exceeds the median.

Should this pattern be treated as a universal theorem for every dataset?

Symmetry without one peak

A symmetric histogram has two equal peaks on opposite sides of its center.

Must the mode equal the mean?

Choosing the center

Synthetic response times have a pronounced right tail.

Which center is less sensitive to unusually slow responses?

A shift and shape

Every file size increases by 5 KB.

Does adding the same constant change the direction of skew?

Toward later models

One dataset is roughly bell-shaped; another has a strong right tail.

Which looks more compatible with a symmetric normal model?